Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

GTIIT
Calculus 1 - M
Winter Semester - 2024

Lecture 22 - December 25th

SEQUENCES AND SERIES

Proposition 1. Let $r \in \mathbb{R}$, then the sequence r^n is convergent if $-1 < r \leq 1$ and divergent in all other cases.
If $-1 < r \leq 1$:

- $\lim_{n\to\infty} r^n = 0 \; \forall -1 < r < 1$
- $\lim_{n\to\infty} 1^n = 1$

Proof. 1) The statement for $r > 0$ is a consequence of the corresponding exponencial function.
2) In case $r < 0$, take $|r^n|$, apply the previous result and the properties.

- Convergence in case $0 < r < 1$:

$\lim_{n\to\infty} r^n = \lim_{x\to\infty} r^x = 0$

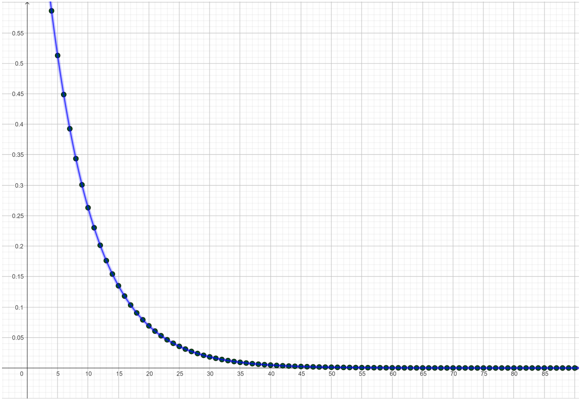
- Convergence in case $-1 < r < 0$: $\lim_{n\to\infty} r^n = 0$,

because of Theorem 5 of Lecture 21- 1st Parte:

$\boxed{\text{If } \lim_{n\to\infty} |a_n| = 0 \Longrightarrow \lim_{n\to\infty} a_n = 0}$

CONVERGENCE IN CASE $0 < r < 1$: $\lim_{n\to\infty} r^n = 0$

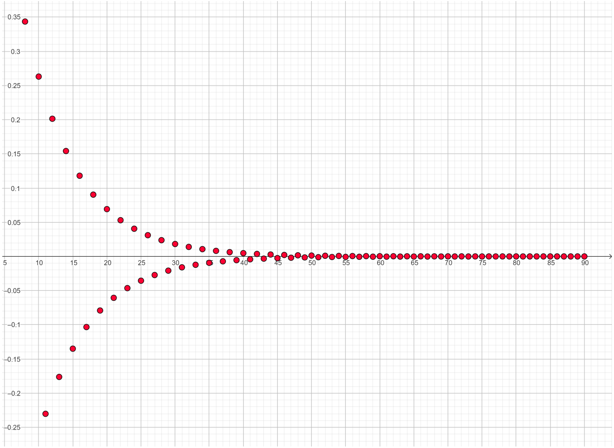
- Ex $\{(\frac{1}{2})^n\} = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32} \cdots\} \Longrightarrow \lim_{n\to\infty}(\frac{1}{2})^n = 0$



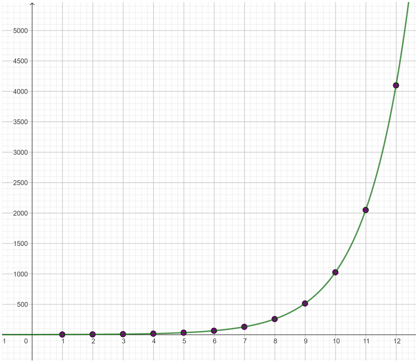
$\lim_{n\to\infty}(\frac{7}{8})^n = 0$

Convergence in case $-1 < r < 0$: $\lim_{n\to\infty} r^n = 0$

- Ex $\{(-\frac{1}{2})^n\} = \{-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32} \cdots\} \Longrightarrow \lim_{n\to\infty}(-\frac{1}{2})^n = 0$



- $\lim_{n\to\infty}(-\frac{7}{8})^n = 0$



$$\lim_{n\to\infty} r^n = \lim_{x\to\infty} r^x = +\infty : r > 1$$

Proof. Given any $M > 0$ take $N > \log_r(M)$ then for any $n > N$:

$$\boxed{n > N \Longrightarrow r^n > r^N > r^{\log_r(M)} = M \Longrightarrow r^n > M \checkmark}$$

Divergence of r^n for $r < -1$: The limit $\lim_{n\to\infty} r^n$ does not exists

Proof. • Suppose $\lim_{n\to\infty} r^n = L \in \mathbb{R} \implies \lim_{n\to\infty} |r^n| = |L|$, but this is not possible for $|r| > 1$:

$$\lim_{n\to\infty} |r^n| = \lim_{n\to\infty} |r|^n = \lim_{x\to\infty} |r|^x = \infty, \text{ contradiction.}$$

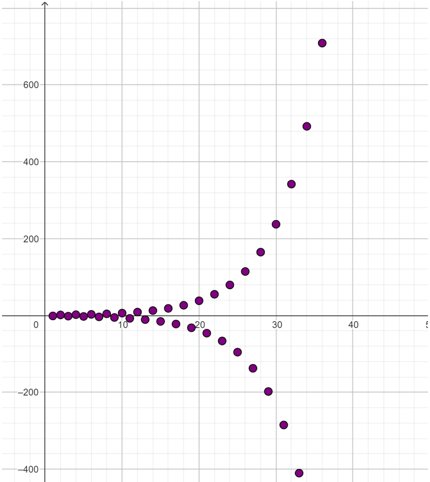
• Suppose $\lim_{n\to\infty} r^n = \infty$:
This is also a contradiction, since there are infinitely many terms $r^n < 0$.

• Suppose $\lim_{n\to\infty} r^n = -\infty$:
Once more, this is also a contradiction, since there are infinitely many terms $r^n > 0$.

So, $\boxed{r^n \text{ does not tend to } L\text{, nor to } +\infty\text{, nor to } -\infty \ \checkmark}$.

DIVERGENCE OF r^n FOR $r < -1$

- r^n does not tend to $+\infty$, neither to $-\infty$



Ex. $(-2)^n = \{-2, 4, -8, 16, -32, 64, -128, 256, \cdots\}$

Monotonic Sequences.

2. Definition.

- A sequence is $\{a_n\}$ is called increasing if $a_n < a_{n+1}$ for all $n \geq 1$, that is, $a_1 < a_2 < a_3 < \cdots$.

- A sequence is $\{a_n\}$ is called decreasing if $a_n > a_{n+1}$ for all $n \geq 1$, that is, $a_1 > a_2 > a_3 > \cdots$.

- A sequence is $\{a_n\}$ is called monotonic if it is either increasing or decreasing.

3. Lemma.

• If $f(x)$ is an increasing function on $[1,\infty)$ and $f(n)=a_n$, then the sequence is $\{a_n\}$ is increasing.

• If $f(x)$ is an decreasing function on $[1,\infty)$ and $f(n)=a_n$, then the sequence is $\{a_n\}$ is decreasing.

Proof. If $f(x)$ is a increasing function on $[1,\infty)$ and $f(n)=a_n$, then

$$\underbrace{f(n)}_{=a_n} < \underbrace{f(n+1)}_{=a_{n+1}} \Longrightarrow a_n < a_{n+1}$$

so, the sequence is $\{a_n\}$ is increasing ✓

Analogously, in the decreasing case.

4. Example. Show that the sequence

$a_n = \frac{n}{n^2+1}$ is decreasing.

- $a_1 = \frac{1}{2} > a_2 = \frac{2}{5} > a_3 = \frac{3}{10} > a_4 = \frac{4}{17} > a_5 = \frac{5}{26} > a_6 = \frac{6}{37} \cdots$

$$0.5 > 0.4 > 0.3 > 0.24 > 0.19 > 0.16 \cdots$$

- Solution 1. Show that $a_{n+1} < a_n$, that is,

$$\frac{(n+1)}{(n+1)^2+1} < \frac{n}{n^2+1}$$

This is equivalent to the one we get taking reciprocals:

$$\frac{(n+1)}{(n+1)^2+1} < \frac{n}{n^2+1} \iff (n+1)(n^2+1) < n[(n+1)^2+1]$$

$$\iff n^3 + n^2 + n + 1 < n^3 + 2n^2 + 2n \iff 1 < n^2 + n \checkmark$$

Monotonic Sequences

Solution 2. Consider $f(x) = \dfrac{x}{x^2+1} \implies f(n) = a_n$, and show that f is decreasing:

$$f'(x) = \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2} < 0 \quad \forall\, x > 1$$

Hence, f is decreasing on $(1, \infty)$ and so, $\{a_n\}$ is decreasing✓

Bounded Sequences

5. Definition

- A sequence $\{a_n\}$ is bounded above if there is a number M such that $a_n \leq M$ for all $n \geq 1$.

- A sequence $\{a_n\}$ is bounded below if there is a number m such that $m \leq a_n$ for all $n \geq 1$.

- If $\{a_n\}$ is bounded above and below, then $\{a_n\}$ is a bounded sequence.

6. Examples.

- The sequence $\{a_n = n\}$ is bounded below by $m = 0 < a_n \ \forall n$ but not above, since $\lim_{n\to\infty} n = \infty$ ✓

- The sequence $\{a_n = (-1)^n\}$ is bounded (above and below) by

$$-1 \leq (-1)^n \leq 1 \ \forall n$$

but it is not convergent.

- The sequence $\{a_n = \frac{n}{n+1}\}$ is bounded (above and below) by

$$0 < \frac{n}{n+1} < 1 \ \forall n$$

- The sequence $\{a_n = \frac{n}{n+1}\}$ is increasing and convergent:

$$\lim_{n\to\infty} \frac{n}{n+1} = 1 \checkmark$$

Completeness Axiom

7. a) Existence of the Supremum.

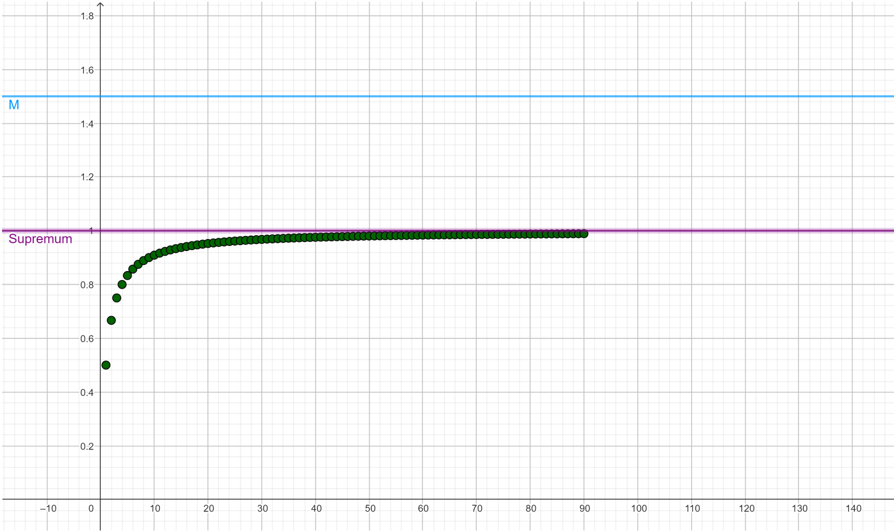
Theorem. If S is nonempty set of real numbers, that is bounded above, then S has a lowest upper bound L, called the Supremum.

- That is, among all the upper bounds of S, there exists the smallest upper bound.

- The Completeness Axiom is an expression of the fact that there is no gap or hole in $\mathbb{R}$, the real number line.

Existence of the Supremum.

Among all the upper bounds of $\{a_n\}$, there is the smallest upper bound called the supremum



Completeness Axiom

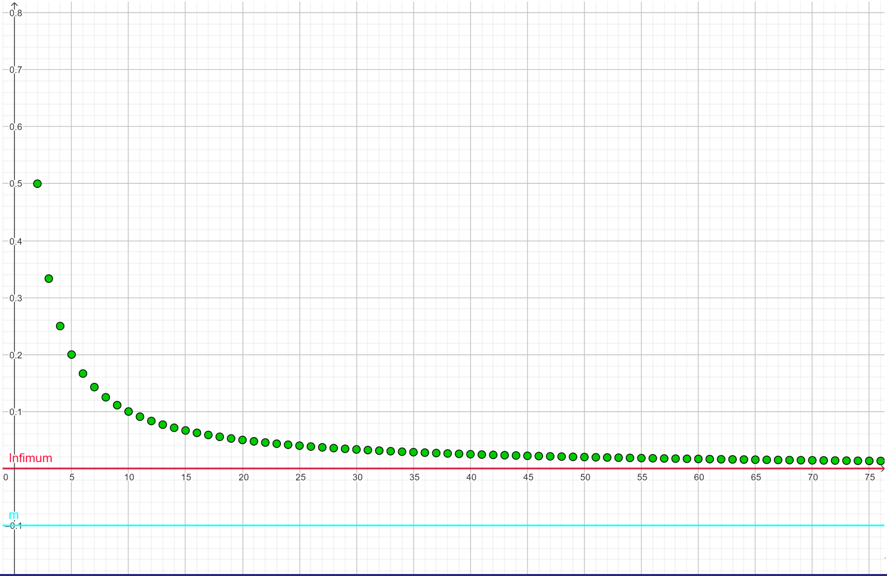
7. b) Existence of the Infimum.

Theorem. If S is nonempty set of real numbers, that is bounded below, then S has a largest lower bound I, called the Infimum.

- That is, among all the lower bounds of S, there exists the largest lower bound.

Existence of the Infimum.

Among all the lower bounds of $\{a_n\}$, there is the largest lower bound called the infimum



8. Monotonic Sequence Theorem.

Every bounded, monotonic sequence is convergent.

This means that, if $\{a_n\}$ is a bounded, increasing or decreasing sequence, then it has a limit, so there exists $L \in \mathbb{R}$:

$$\lim_{n\to\infty} a_n = L$$

- If $\{a_n\}$ is increasing, the limit $L =$ Supremum of S
- If $\{a_n\}$ is decreasing, the limit $L =$ Infimum of S ✓

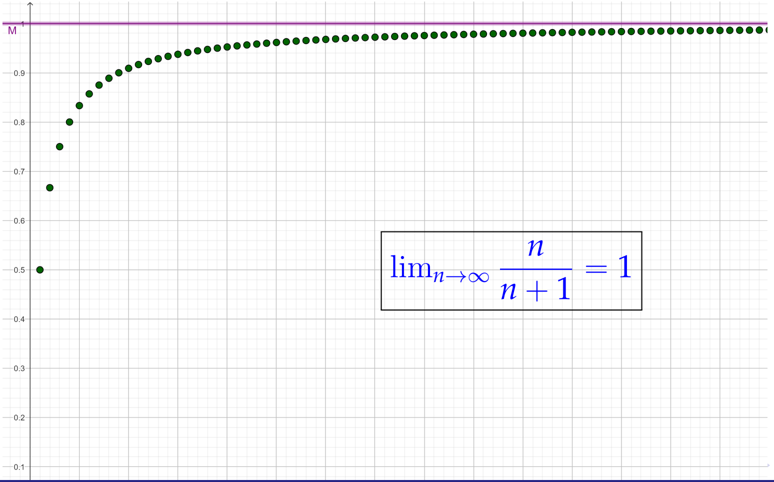
Monotonic Sequences.

Example. The sequence $\lim_{n\to\infty} \frac{n}{n+1} = 1$, then, its supreme is $L = 1$,

- $a_n = \frac{n}{n+1} < 1 \ \forall n$, so, $\{a_n\}$ is bounded by $M = 1$

- The Supremum of $S = \{\frac{n}{n+1}\}$ is $L = 1$ because of the theorem, since the limit is $L = 1$ ✓

Existence of the Supremum.



An Example of a Recurrent Sequence

9. Example. Show that a_n is an increasing and bounded sequence and find its limit:

$$a_1 = 2,\ a_{n+1} = \frac{1}{2}(a_n + 6) :\ n \geq 1$$

Solution. • $a_1 = 2, \quad a_2 = \frac{1}{2}(2+6) = 4, \quad a_3 = \frac{1}{2}(4+6) = 5,$

$a_4 = \frac{1}{2}(5+6) = \frac{11}{2}, \quad a_5 = \frac{1}{2}(\frac{11}{2}+6) = 5.75, \quad a_6 = 5.875,$

$a_7 = 5.94,\ a_8 = 5.97,\ a_9 = 5.98,\ a_{10} = 5.99$

a_n is increasing

- Prove by induction that a_n is increasing, that is, we want to show $a_n < a_{n+1} : n \geq 1$

  - Case $n = 1$: $a_1 = 2 < 4 = a_2$ ✓

- Inductive Step: Suppose that $a_n < a_{n+1}$ is true

$$\implies a_n + 6 < a_{n+1} + 6$$

$$\implies \underbrace{\tfrac{1}{2}(a_n + 6)}_{=a_{n+1}} < \underbrace{\tfrac{1}{2}(a_{n+1} + 6)}_{=a_{n+2}} \checkmark$$

$$\implies a_{n+1} < a_{n+2} \checkmark$$

So, the inequality is true for all $n \in \mathbb{N}$ ✓

$a_n = \frac{1}{2}(a_n + 6)$ is bounded: $2 \leq a_n < 6\ \forall n$

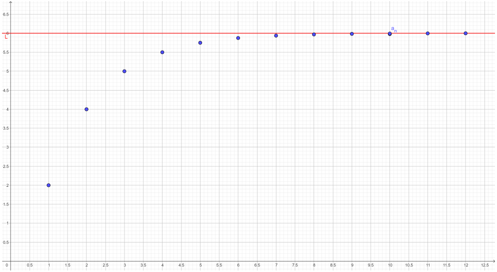
- $a_1 = 2$ and $a_1 < a_n\ \forall n$ because a_n is increasing, then $2 \leq a_n\ \forall n$

- Prove by induction that $a_n < 6\ \forall n$:
  - Case $n = 1$: $a_1 = 2 < 6$ ✓
  - Inductive Step: that is, suppose that $a_n < 6$ for any n, and show $a_{n+1} < 6$:

$$a_n < 6 \implies a_n + 6 < 12 \implies \underbrace{\frac{1}{2}(a_n + 6)}_{=a_{n+1}} < \underbrace{\frac{1}{2}\ 12}_{=6} \implies a_{n+1} < 6.\ \checkmark$$

Hence, $a_n < 6\ \forall n$ ✓

If a_n is increasing and bounded, then it converges



$$\boxed{L = \lim_{n\to\infty} \frac{1}{2}(a_n + 6) = 6}$$

Find $L = \lim_{n\to\infty} a_n$

- Theorem: If $\{a_n\}$ is increasing and bounded, then there exists $\lim_{n\to\infty} a_n = L \in \mathbb{R}$

- Recall $\boxed{a_1 = 2, a_{n+1} = \frac{1}{2}(a_n + 6) : \ n \geq 1}$

- $\boxed{\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} a_{n+1}}$, since this is the same sequence, except for the 1st term, and it goes one term forward

$$\boxed{L} = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \frac{1}{2}(a_n + 6) \ \boxed{= \frac{1}{2}(L+6)}$$

$$\Longrightarrow \boxed{L = \frac{1}{2}(L+6) \Longrightarrow L = 6 \ \checkmark}$$

Series

A series is the sum of the elements of a sequence: $\sum_{k=1}^{\infty} a_k$.

Example: Given $\left\{\frac{1}{2^n}\right\}_{n=1}^{\infty}$: $\{a_n\} = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \cdots\right\}$

Consider the new sequence, the series, obtained by the sum of the first n terms:

$$S_1 = a_1 = \frac{1}{2},\ S_2 = a_1 + a_2 = \frac{3}{4},\ S_3 = a_1 + a_2 + a_3 = \frac{7}{8}$$

$$S_4 = \sum_{k=1}^{4} a_k = \frac{15}{16},\ S_5 = \sum_{k=1}^{5} a_k = \frac{31}{32},\ \cdots, S_{10} = \sum_{k=1}^{10} a_k = \frac{1023}{1024}$$

$$S_{20} = \sum_{k=1}^{20} a_k = 0.99999904\ldots \qquad S_{30} = \sum_{k=1}^{30} a_k = 0.9999999990686\ldots$$

n	$a_1 + \cdots + a_n$	$\sum_{k=1}^{n} 1/2^k$
1	$\frac{1}{2}$	0.5
2	$\frac{1}{2} + \frac{1}{4}$	0.75
3	$\frac{1}{2} + \frac{1}{4} + \frac{1}{8}$	0.875
4	$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$	0.9375
5	$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$	0.96875
10	$\frac{1}{2} + \cdots + \frac{1}{2^{10}}$	0.999023...
20	$\frac{1}{2} + \cdots + \frac{1}{2^{20}}$	0.999999046...
30	$\frac{1}{2} + \cdots + \frac{1}{2^{30}}$	0.999999999...

We say that the series converges to $L = 1$ and write:

$$\boxed{\sum_{k=1}^{\infty} \frac{1}{2^k} = 1}$$

- The table shows that as we add more and more terms, these partial sums become closer and closer to 1.

- In fact, by adding sufficiently many terms of the series we can make the partial sums as close as we like to 1.

- So it seems reasonable to say that the sum of this infinite series is 1 and to write

$$\boxed{\sum_{k=1}^{\infty} \frac{1}{2^k} = \lim_{n\to\infty}\left(\sum_{k=1}^{n} \frac{1}{2^k}\right) = 1}$$

SERIES

10. Definition. • A series is the sum of the elements of a sequence $\sum_{k=1}^{\infty} a_k$.

• The series is called convergent if the limits of the partial sums exists, that is, is the sum up to n, has a finite limit $L \in \mathbb{R}$ as $n \to \infty$:

$$\boxed{\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} \left(\sum_{k=1}^{n} a_k \right) = L}$$

• In other case, the series is called divergent.

Geometric Series

11. A Geometric Series is obtained by adding powers of $r \in \mathbb{R}$

$$\boxed{\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} : \quad -1 < r < 1}$$

- The Geometric Series converges if $-1 < r < 1$ and it diverges if $|r| \geq 1$

Ex. For $r = \frac{1}{2} < 1$, $\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1-r} = \frac{1}{1-\left(\frac{1}{2}\right)} = 2$ ✓

Proof: Compute the partial sum $S_n = \sum_{k=0}^{n} r^k$ and show that they have a limit only for $-1 < r < 1$:

$$(1-r)\left(\sum_{k=0}^{n} r^k\right) = (1-r)\left(1 + r + r^2 + \ldots + r^n\right)$$

$$= 1 + r + r^2 + \ldots + r^n$$

$$-r - r^2 + \ldots - r^n - r^{n+1} \quad \boxed{= 1 - r^{n+1}}$$

$$\boxed{\Longrightarrow \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{(1-r)}}$$

$$\Longrightarrow \sum_{k=0}^{\infty} r^k = \lim_{n\to\infty} \sum_{k=0}^{n} r^k = \lim_{n\to\infty} \left[\frac{1 - \overbrace{r^{n+1}}^{\to 0}}{(1-r)}\right] \quad \boxed{= \frac{1}{1-r} : \ -1 < r < 1}$$

12. Example. Write $2.3\overline{17}$ as a quotient of integers.

Solution. The given is 2.3 plus a pure periodic number

$$\boxed{2.3\overline{17}} = 2.3 + 0.017 + 0.00017 + \ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \ldots$$

$$= 2.3 + \frac{17}{10^3}\left[1 + \tfrac{1}{100} + \tfrac{1}{100^2} + \ldots\right]$$

$$= 2.3 + \frac{17}{10^3} \cdot \left(\sum_{n=0}^{\infty} \left(\tfrac{1}{10^2}\right)^n\right) = 2.3 + \frac{17}{10^3}\left(\frac{1}{1-(\frac{1}{100})}\right)$$

$$\frac{23}{10} + \frac{17}{10^3}\left(\frac{1}{\frac{99}{100}}\right) = \frac{1147}{495} \checkmark$$

13. Theorem. If the series $\sum_{n=0}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$.

- This means that, if the sequence $\{a_n\}$ doesn't have a limit or $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=0}^{\infty} a_n$ is divergent.

- Example a) The series $\sum_{n=0}^{\infty} (-1)^n$ diverges: $\lim_{n\to\infty} (-1)^n$ does not exists and

$$\sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + 1 - 1 + 1 \ldots \text{is divergent}$$

- Example b) The series $\sum_{n=0}^{\infty} \frac{3n^2+4}{2n^2+n+1} = \infty$, so it is divergent:

$$\lim_{n\to\infty} \frac{3n^2+4}{2n^2+n+1} = \frac{3}{2} \neq 0$$

It diverges since the sum has infinitely many terms close to $3/2$

NECESSARY CONDITION FOR CONVERGENCE: THE RECIPROCAL IS NOT TRUE

14. Observation. The fact $\lim_{n\to\infty} a_n = 0$ does not imply (we can not conclude) that the series $\sum_{n=0}^{\infty} a_n$ converges.

The necessary condition is not enough to guarantee the convergence.

- Example c) $\lim_{n\to\infty} \frac{1}{n} = 0$ but the series $\sum_{n=0}^{\infty} \frac{1}{n}$ diverges:

$$\int_1^{\infty} \frac{1}{x} = \lim_{t\to\infty} \ln(t) - 1 = \infty$$

Proof. $\boxed{\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \dots}$

$$= 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{2 \ terms} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{2^2 \ terms} + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16}\right)}_{2^3 \ terms} + \dots$$

$$\geq 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{4} + \frac{1}{4}\right)}_{2 \ terms} + \underbrace{\left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)}_{2^2 \ terms} + \underbrace{\left(\frac{1}{16} + \frac{1}{16} + \dots + \frac{1}{16}\right)}_{2^3 \ terms} + \dots$$

$$= 1 + \frac{1}{2} + \frac{2}{4} + \frac{4}{8} + \frac{8}{16} + \dots =$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots = +\infty \ \checkmark$$

15. Theorem. Operations with Series

If $c \in \mathbb{R}$, $\sum_{n=1}^{\infty} a_n = L_1$ and $\sum_{n=1}^{\infty} b_n = L_2$ are convergent series, that tend to L_1 and $L_2 \in \mathbb{R}$, respectively, then the following operations are convergent series and:

1) $\boxed{\sum_{n=1}^{\infty} \left(a_n + b_n\right) = \left(\sum_{n=1}^{\infty} a_n\right) + \left(\sum_{n=1}^{\infty} b_n\right) = L_1 + L_2}$

2) $\boxed{\sum_{n=1}^{\infty} \left(a_n - b_n\right) = \left(\sum_{n=1}^{\infty} a_n\right) - \left(\sum_{n=1}^{\infty} b_n\right) = L_1 - L_2}$

3) $\boxed{\sum_{n=1}^{\infty} \left(c \cdot a_n\right) = c \cdot \left(\sum_{n=1}^{\infty} a_n\right) = c \cdot L_1}$

1. $\int_1^\infty \frac{1}{x^p}dx = -\lim_{t\to\infty} \frac{1}{(p-1)x^{p-1}}\Big|_1^t = \frac{1}{(p-1)}$ for $p > 1$
2. $\int_1^\infty \frac{1}{x}dx = \lim_{t\to\infty} \ln(x)\Big|_t^1 = +\infty$

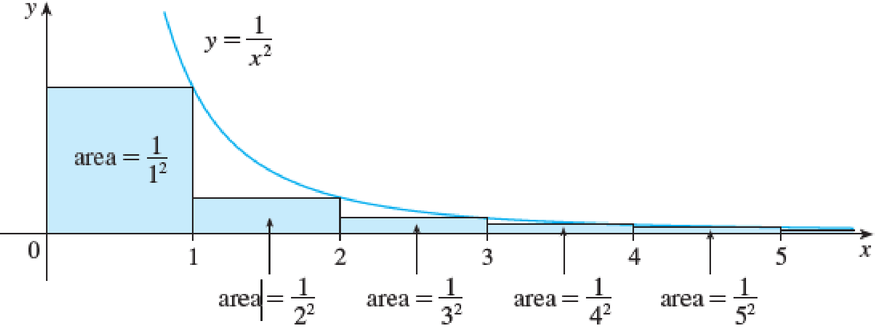
$\Rightarrow$ The integral on $[1,\infty)$ is convergent if and only if $p > 1$ ✓

3. $\int_0^1 \frac{1}{x^p}dx = \lim_{t\to 0^+} \frac{x^{1-p}}{(1-p)}\Big|_t^1 = \frac{1}{(1-p)}$ *for* $p < 1$
4. $\int_0^1 \frac{1}{x}dx = \lim_{t\to 0^+} \ln(x)\Big|_t^1 = -\infty$

$\Longrightarrow$ The integral on $[0,1]$ is convergent if and only if $p < 1$ ✓

17. Integral Test

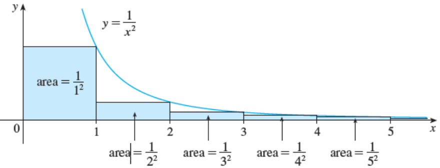
Idea: compare the series=sum of the areas of the rectangles with the area of the corresponding area under the curve $f(x)$ if $f(n) = a_n$

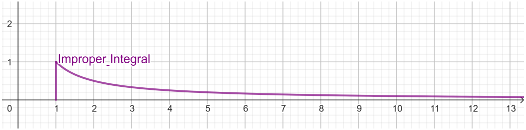


$$\int_1^\infty \frac{1}{x^2}\,dx \text{ is convergent} \iff \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ is convergent}$$

Compare a Series with an Improper Integral

$$\int_1^{\infty} \frac{1}{x^2}dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^2}dx = \lim_{t\to\infty}\left[-\frac{1}{x}\Big|_1^t\right] = \lim_{t\to\infty}\left[-\underbrace{\left(\frac{1}{t}\right)}_{\to 0}+1\right] = 1$$





$\Longrightarrow$ The improper integral is convergent

n	$\sum_{k=1}^{n} 1/k^2$
5	$1.4636\ldots$
10	$1.5498\ldots$
50	$1.6251\ldots$
100	$1.6350\ldots$
500	$1.6429\ldots$
1000	$1.6439\ldots$
5000	$1.6447\ldots$
10^4	$1.6448\ldots$

The table of approximate values suggests that the partial sums have a limit as $n \to \infty$ and in fact, the series is convergent:

$$\boxed{\lim_{N\to\infty} \sum_{n=1}^{N} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.6449}$$

- The base of each rectangle is an interval of length 1; the height is equal to the value of the function $f(x) = 1/x^2$ at the right endpoint of the interval.
- The rectangles lie below the curve, since f is decreasing.
- The sum of the areas of the rectangles is

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

- If we exclude the 1st rectangle, the total area of the remaining rectangles is smaller than the area under the curve $y = 1/x^2$ for $x \geq 1$, which is the value of the convergent improper integral:

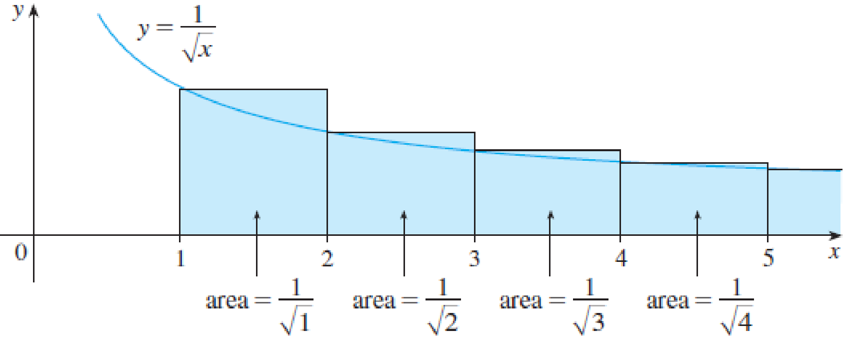
$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} \leq \int_1^{\infty} 1/x^2 \, dx = 1}$$

- So all the partial sums are less than 2:
$$\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots=\sum_{n=1}^{\infty}\frac{1}{n^2}<\frac{1}{1^2}+\int_1^{\infty}1/x^2\,dx=2.$$

- Thus the partial sums are bounded. We also know that the partial sums are increasing, because all the terms are positive.

- Therefore the partial sums converge by the Monotonic Sequence Theorem, and so the series is convergent.

- To get to $=\sum_{n=1}^{\infty}\frac{1}{n^2}=\pi^2/6$ is an exercise.



$$\int_1^\infty \frac{1}{\sqrt{x}}dx = \lim_{t\to\infty}\int_1^t \frac{1}{\sqrt{x}}dx = \lim_{t\to\infty}\left[2x^{1/2}\Big|_1^t\right] = \lim_{t\to\infty}\left[\underbrace{(2t^{1/2})}_{\to+\infty} - 2\right] = +\infty$$

$$\Longrightarrow \int_1^\infty \frac{1}{\sqrt{x}}\,dx \text{ is divergent} \iff \sum_{n=1}^\infty \frac{1}{\sqrt{n}} \text{ is divergent} \checkmark$$

n	$\sum_{k=1}^{n} 1/\sqrt{k}$
5	3.2...
10	5.0...
50	12.7...
100	18.6...
500	43.3...
1000	61.8...
5000	139.9...

The table of approximate values suggests that the partial sums don't have a limit as $n \to \infty$ and in fact, the series is divergent:

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \infty}$$

- The base of each rectangle is an interval of length 1; the height is equal to the value of the function $f(x) = 1/\sqrt{x}$ at the left endpoint of the interval.
- The rectangles lie above the curve.
- The sum of the areas of the rectangles is

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \ldots = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

- The total area of the rectangles is greater than the area under the curve $y = 1/\sqrt{x}$ for $x \geq 1$, which is the value of the divergent improper integral: $\int_1^{\infty} \frac{1}{\sqrt{x}}\, dx = \infty$.

  So, the series is divergent ✓

Theorem. Integral Test.

Let f be continuous, positive, decreasing on $[1, \infty)$ such that $f(n) = a_n$. Thus, the series $\sum_{k=1}^{\infty} a_n$ is convergent if and only if the improper integral $\displaystyle\int_1^{\infty} f(x)\,dx$ is convergent.

In other words,

1. If $\displaystyle\int_1^{\infty} f(x)\,dx$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is convergent.
2. If $\displaystyle\int_1^{\infty} f(x)\,dx$ is divergent, then $\sum_{k=1}^{\infty} a_k$ is divergent.

- The starting term may be any $k_0 \in \mathbb{N}$, not necessarily $k = 1$, and so, the interval for the corresponding improper integral is (k_0, ∞) ✓.
- It is enough that f is decreasing on an interval (c, ∞) for any $c > 0$.

18. Corollary The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if $p > 1$ and divergent if $p \leq 1$ ✓.

Examples: • The series is convergent: $\sum_{n=1}^{\infty} \dfrac{1}{n^2}$ because of the convergence of the integral $\int_1^{\infty} \frac{1}{x^2} dx = 1$.

• The series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n^3}}$ is divergent because of the divergence of the improper integral $\int_1^{\infty} \frac{1}{\sqrt[5]{x^3}} dx = \int_1^{\infty} \frac{1}{x^{3/5}} dx = +\infty : p = \frac{3}{5}$

19. Limit Comparison Test. Suppose $a_n, b_n > 0 \ \forall n$.

a) If $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$ then either both series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent or both series are divergent.

b) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} b_n$ is convergent then the series $\sum_{n=1}^{\infty} a_n$ is also convergent.

c) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} a_n$ is divergent then the series $\sum_{n=1}^{\infty} b_n$ is also divergent.

Limit Comparison Test and Integral Test

Ex. 11. Decide about the convergence or divergence of the following series

a) $\displaystyle\sum_{n=1}^{\infty} \frac{\pi n + 2}{6n^3 - 4}$ b) $\displaystyle\sum_{n=1}^{\infty} \sqrt[4]{\frac{n^3 + 6}{n^6 - 2}}$.

• Suppose we have been able to use the Integral Test to show that a series is convergent.
• Next we want to find an approximation to the sum S of the series. Any partial sum S_n is an approximation to S because $S = \sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} \sum_{k=1}^{n} a_k$.
• How good is such an approximation? To find out, we need to estimate the size of the remainder

• The remainder R_n is the difference between the whole sum and a partial sum, so R_n is the error made when S_n, the sum of the first n terms, is used as an approximation to the total sum:

$$\boxed{R_n = S - S_n = \sum_{k=n+1}^{\infty} a_k = a_{n+1} + a_{n+2} + \ldots}$$

Remainder Estimate for the Series

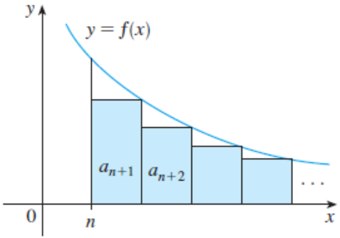
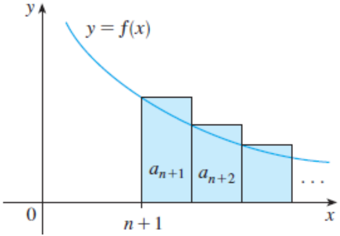


FIGURE 3



Remainder Estimate for the Series

$$\int_{n+1}^{\infty} f(x)\,dx$$

$$\leq R_n = S - \sum_{k=1}^{n} a_k = a_{n+1} + a_{n+2} + a_{n+3} + \ldots$$

$$\leq \int_{n}^{\infty} f(x)\,dx$$

21. Theorem Let f be a continuous, positive, decreasing function on $[1, \infty)$ such that $f(n) = a_n$.
Denote by $S_n = \sum_{k=1}^{n} a_k$. Suppose that the series is convergent, that is, $\sum_{k=1}^{\infty} a_k = S \in \mathbb{R}$.
Thus, there is the following relationship between the remainder R_n and the improper integrals:

$$\boxed{\int_{n+1}^{\infty} f(x)\, dx \leq R_n = S - \sum_{k=1}^{n} a_k \leq \int_{n}^{\infty} f(x)\, dx}$$

Equivalently,

$$\boxed{S_n + \int_{n+1}^{\infty} f(x)\, dx \leq S \leq S_n + \int_{n}^{\infty} f(x)\, dx}$$

22. a) Approximate the series $\sum_{k=1}^{\infty} \frac{1}{k^3}$ by using the sum of the first 10 terms. Estimate the error involved in this approximation.
(b) How many terms are required to ensure that the partial sum S_n is exact to within 5×10^{-4}?

Solution. • $f(x) = \frac{1}{x^3}$ is continuous, decreasing and positive:

- f is decreasing, because $f'(x) = \frac{-3}{x^4} < 0$
- Estimation of the remainder:

$$R_n \leq \int_n^{\infty} \frac{1}{x^3}\, dx = \lim_{t\to\infty} \frac{-1}{2x^2}\Big|_n^t = \lim_{t\to\infty} [-\frac{1}{2t^2} + \frac{1}{2n^2}] \boxed{= \frac{1}{2n^2}}$$

- Estimation of the series and remainder:

$$\sum_{k=1}^{\infty} \frac{1}{k^3} \approx S_{10} = \frac{1}{1^3} + \frac{1}{2^3} + \ldots + \frac{1}{10^3} \approx 1.1975 \text{ and } R_{10} \leq \frac{1}{200} = 0.005$$

Remainder Estimate for the Series

(b) How many terms are required to ensure that the partial sum S_n is accurate to within 5×10^{-4}?

- Use the estimation of the reminader to obtain n to get accuracy to within 5×10^{-4}:

$$R_n \leq \int_n^{\infty} \frac{1}{x^3}\, dx \quad \boxed{= \frac{1}{2n^2} < 5 \times 10^{-4}}$$

$$\iff \quad n^2 > \frac{1}{10^3} = 10^3 \quad \iff \quad \boxed{n > \sqrt{10^3} \approx 31.6 \Longrightarrow n \geq 32 \ \checkmark}$$

Accuracy of 5×10^{-4}

The partial sum S_{32} provides an approximation of $\sum_{k=1}^{\infty} \frac{1}{k^3}$ with the desired accuracy

- $\sum_{k=1}^{32} \frac{1}{k^3} \approx 1.20158364...$ and
- $\sum_{k=1}^{\infty} \frac{1}{k^3} \approx 1.2020569...$ ✓

So its difference is $0.00047...$ less than 0.0005 ✓